MILITARY SPECIFICATION

TEST SET, TRANSISTOR: AN/USM - 136

1. SCOPE

1.1 This specification covers one type of transistor test set used to test transistors, semiconductor diodes and other semiconductor devices. It is capable of testing semiconductor devices in an automatic or manual mode on an "out-of-circuit" or "in-circuit" basis and makes use of punched card programming.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-T-945	Test Equipment for Use with Electronic Equipment; General Specification for
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment
MIL-C-9436	Chamber, Sand and Dust Testing
MIL-F-13926	Fire-control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-F-14252	Fire Control Materiel: General Specification Governing the Design and Inspection of
MIL-I-45208	Inspection Requirements: General Specification for
MIL-S-55203	Sub-assemblies, Plug-in, for Transistor Test Set: AN/USM-136
MIL-M-55204	Module, plug-in, Automatic Lead Selector - In-Circuit Checker, for Transistor Test Set: AN/USM-136

Source: https://assist.dla.mil -- Downloaded: 2026-06-26 04:14Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-STD-151	Metals; Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	Definition of, and Basic Requirements for, Enclosures for Electric and Electronic Equipment

DRAWINGS

Munitions Command

A8625349	Assembly, Test Set, Transistor; AN/USM-136
F8625324	Readout Converter
F8625337	h_{FE} (pulse) Base Drive Circuit
F8625338	Scaling Amplifier
F8625340	Comparison Converter
F8625341	Readout Counter
F8625342	Light Driver
D8625285	h_{fe} Base Drive Circuit
F8625449	Automatic Lead Selector - In-Circuit Checker
C8625532	Cross Reference, Electrical Schematics
D8625444	Inspection Test Cards

PACKAGING DATA SHEET

MIL-P-14232/ P8625329	Packaging of Test Set, Transistor: AN/USM-136

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Initial production sample . - Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production.

3.2 Materials. - Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Fabrication. - The transistor test set shall be manufactured in accordance with Drawing A8625349 "Assembly, Test Set, Transistor: AN/USM-136" and all drawings and specifications pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance, and permanence to insure inherent quality and performance capabilities.

3.4 General specifications. - The contractor shall be responsible for adherence to, and compliance with the following specifications: MIL-T-945, MIL-E-5272, MIL-F-13926, and MIL-F-14252, particularly the following paragraphs:

3.4.1 Specification MIL-E-5272

(a) High temperature test, Procedure II

(b) Low temperature test, Procedures I and II

(c) Temperature shock test, Procedure I

(d) Humidity test, Procedure I

(e) Altitude test, Procedure VI

(f) Salt spray test, Procedure I

(g) Sand and dust test, Procedure I

(h) Shock, Procedure V, equipment operation

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.4.2 Specification MIL-F-13926

(a) Order of precedence
(b) Protective finished surfaces
(c) Lubrication

3.4.3 Specification MIL-F-14252

(a) Use of standard items
(b) Precedence of standard drawings
(c) Wired equipment
(d) Workmanship

3.5 Environmental.

3.5.1 Operating temperatures. - The transistor test shall meet all the performance requirements of this specification, when thermally stabilized at ambient temperatures of -20°C (-4°F) to +50°C (+122°F).

3.5.2 Storage temperatures. - The transistor test set shall meet all the performance requirements of this specification upon being thermally stabilized at operating temperature after having been thermally stabilized at ambient temperatures of -62°C (-80°F) and +71°C (+160°F).

3.5.3 Operating altitude. - The transistor test set shall meet all the performance requirements of this specification, when thermally stabilized at a temperature of -20°C (-4°F) and a pressure of 20.58" Hg (10,000 ft. altitude).

3.5.4 Non-operating altitude. - The transistor test shall meet all the performance requirements of this specification after being subjected to a pressure of 3.44" Hg (50,000 ft. altitude) and a temperature of -54°C (-65°F). The performance test shall be conducted at the operating altitude parameters of 3.5.3.

3.5.5 Temperature shock. - The transistor test set shall be capable of meeting all the performance requirements of this specification upon return to room temperature of +25°C (77°F) after being subjected to a temperature shock test, the limits of which are -40°C (-40°F) and +85°C (+185°F).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.5.6 Shock. - The transistor test set shall be capable of withstanding 18 shocks of 15 G's acceleration for a duration of 11 plus or minus 1 millisecond. The transistor test set shall not suffer damage or subsequently fail to provide the performance specified in this specification.

3.5.7 Vibration. - The transistor test set shall be capable of meeting the performance requirements of this specification after being subjected to the following vibration tests:

(a) A simple harmonic motion of 0.06 inch at a frequency varying from 10 to 33 cycles per second and back to 10 cycles per second.

(b) A simple harmonic motion of 0.03 inch at a frequency varying from 10 to 55 cycles per second and back to 10 cycles per second.

(c) Vibration at the four most severe resonant frequency points.

3.5.8 Humidity. - The transistor test set shall meet all the performance requirements of this specification after being subjected to ten cycles of the humidity test described in paragraph 4.9.1.3.8 of this specification.

3.5.9 Sand and dust. - The transistor test set shall not suffer any internal damage and shall meet all the performance requirements of this specification, after exposure with the cover on, to a sand and dust density of a 1 to 0.5 grams per cubic foot.

3.5.10 Splashproof. - The transistor test set shall not suffer any internal damage and shall meet all the performance requirements of this specification after exposure with the covers on, to the splashproof tests specified in Standard MIL-STD-108.

3.5.11 Salt spray. - The transistor test set shall meet all the performance requirements of the specification after being subjected to a salt spray for 50 hours. The requirements for the salt spray test are set up in Method 811 of Standard FED-STD-151.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.6 Performance requirements

3.6.1 Subassemblies. - The following subassemblies of the transistor test set shall meet the requirements of their respective individual specifications as listed below:

	Subassemblies	DWG. Number	Specification
(a)	Plug-in Boards		MIL-S-55203
	(1) Readout converter	F8625324	
	(2) h_{FE} (pulse) Base drive circuit	F8625337	
	(3) Scaling amplifier	F8625338	
	(4) Comparison converter	F8625340	
	(5) Readout counter	F8625341	
	(6) Light driver	F8625342	
	(7) h_{fe} Base drive circuit	D8625285	
(b)	Automatic lead selector In-Circuit checker	F8625449	MIL-M-55204

3.6.2 Circuitry. - The transistor test set circuitry shall be in accordance with electrical schematics specified on Drawing C8625532, "Cross Reference". The transistor test set should be capable of performing the following tests on all presently known transistors and semiconductor devices that can be programmed within the performance envelope limits of this equipment on an "out-of-circuit" basis. In addition, the transistor test set should have the capability of checking transistors and diodes "in-circuit" where feasible.

3.6.3 Warm-up. - The transistor test set shall be fully operable within 5 minutes after the main power is turned on.

3.6.4 Readout.

3.6.4.1 Projection digital voltmeter. - The transistor test set shall include a two significant figure digital voltmeter with an input range of 0.1 V. to 9.9 volts and an output presentation having a plus or minus 2 percent full scale, plus or minus 1 count accuracy, and which uses projection readout. The two significant figure projection readout shall be centered on the readout screen and the system shall include by means of switchable shadow-lights a four (4) place programmable decimal to permit the following readout set-ups:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(a) 0.00
(b) 0.0
(c) 00
(d) 000

An "Off-Scale" indicator light shall appear on the readout screen, when the digital voltmeter output exceeds the count of 99. Results of any programmed test measurement shall be available as a projection readout while in the test position. A front panel switch shall be provided to allow the projection readout of the programmed acceptance limit while in the test position.

3.6.4.2 Scaling amplifier. - A scaling amplifier in conjunction with the programming relays shall extend the range of the projection digital voltmeter below the minimum input value of 0.1 volt, and enable the input signal to be subjected to a gain of X10, X100, and X1,000. The gain accuracy for each range shall be plus or minus 2 percent full scale.

3.6.4.3 Indicators. - A group of lamps and a projector shall indicate the condition of the test set and the semiconductor under test. These shall be as follows:

(a) The twelve (12) digit projector shall indicate the TEST NUMBER.

(b) The yellow READY light shall indicate that both the main test stepper switch and automatic lead selector stepper switch are in the "ready" position.

(c) The green GO lamp indicates that the semiconductor under test has not failed to be within any of the programmed test acceptance limits for out-of-circuit tests or has not failed the in-circuit check.

(d) The red NO-GO lamp indicates that the semiconductor under test fails to be within any one of the programmed test acceptance limits for out-of-circuit tests or fails the in-circuit checks. Operation of the NO-GO light is a result of energizing relay K1311. This relay action also stops the test sequence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(e) The red OPEN lamp indicates that in the forward conduction test of the automatic lead selector - in-circuit checker, a junction in the semiconductor under test appears as not less than 250 ohms for germanium or not less than 500 ohms for silicon, when tested in the LEADS UNKNOWN mode of operation.

(f) The red NO GAIN lamp indicates that in a forward vs. reverse beta check of the automatic lead selector - in-circuit checker, the transistor under test either will not amplify for any reason, has a gain of less than 8, or that the difference between the forward and reverse beta is less than 50 percent, when tested in the LEADS UNKNOWN mode of operation.

(g) The yellow SHORT OR LOW IMPEDANCE lamp indicates that in the reverse conduction test by the automatic lead selector - in-circuit checker, a junction in the semiconductor under test appears shunted by less than 250 ohms for germanium or less than 500 ohms for silicon, when tested in the LEADS UNKNOWN mode of operation.

(h) The blue lamp indicates that AC power is ON.

3.6.5 Test routine. - A twelve step test routine shall be provided in the equipment in the following order:

1. h_{fe} vs. Low limit
2. h_{fe} vs. High limit
3. h_{FE} (pulse) vs. Low limit
4. h_{FE} (pulse) vs. High limit
5. h_{FE} vs. Low limit
6. h_{FE} vs. High limit
7. I_{fwd} vs. Low limit (a current for an applied voltage test)
8. I_{cbo} vs. High limit (a current for an applied voltage test)
9. I_{BV} or I_{rev} vs. High limit (a current for an applied voltage test)
10. V_Z vs. Low limit (a voltage for an applied current test)
11. V_Z vs. High limit (a voltage for an applied current test)
12. Z_Z vs. High limit (AC voltage read as impedance for an applied alternating current superimposed on a DC)

The above test routine may be preceded by the use of automatic lead selection by placing the LEADS switch into the UNKNOWN position. The automatic lead selector - in-circuit checker assembly, F8625449 is fully specified in MIL-M-55204. When LEADS KNOWN is selected the OPEN and SHORT test feature of the automatic lead selector is not obtainable for the semiconductor under test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.6.6 Programming.

3.6.6.1 Test set-up. - Choice of which test, test conditions, and acceptance limits shall be capable of being programmed into the transistor test set by means of a punched program card. The punched program card shall be capable of being read by a card reader on the front of the transistor test set.

3.6.6.2 Test operation. - The sequency of test operation is initiated by momentarily pressing the TEST button. The performance of the tests programmed on the punched card shall be either by automatic stepping or manual stepping.

3.6.6.2.1 Automatic stepping. - The automatic stepping through the 12 positions which include the specific tests programmed on the punched cards shall require setting the AUTO-MANUAL-ADVANCE switch to the AUTO position and activiation of the TEST button to begin the test sequence. Due to the stepping speed, only Go, No-Go indications need be considered.

3.6.6.2.1.1 Test time.

(a) Automatic Lead Selection - in-circuit test time is 3 seconds maximum.

(b) Test time for programmed semiconductor tests is 6 second maximum.

3.6.6.2.2 Manual stepping. - The manual stepping through any or all of the 12 positions which include the specific tests programmed on the punched cards shall require setting the AUTO-MANUAL-ADVANCE switch to the MANUAL position, activation of the TEST button to establish a start capability and then momentary pressing down the AUTO-MANUAL-ADVANCE switch to the ADVANCE position for each test position.

3.6.6.3 Programming the punched cards. - The punched program card shall be capable of programming the following tests within the limits of operation of the individual test circuits, power supplies, and acceptance limits:

(a) Transistors, PNP or NPN

either (1) Small signal Beta test (h_{fe}) against high and low limits.
or (2) Large signal Beta test (h_{FE}) (pulse) against high and low limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

or (3) Large signal DC Beta (h_{FE}) against high and low limits.
and (4) I_{cbo} test against a high limit.
and (5) I_{BV} test against a high limit.

(b) Semiconductor diodes

(1) Forward current (I_{fwd}) against a low limit.
and (2) Reverse current (I_{rev}) against a high limit.

(c) Zener diodes

(1) Zener voltage against high and low limits.
and (2) Zener impedance against a high limit.

3.6.6.4 Programmed acceptance limits. - The programmed acceptance limits for any punched card can be programmed to four (4) separate values within the ninety-nine (99) increments of 01 and through 99 and shall have an accuracy of plus or minus 2 percent.

3.6.6.5 Programmable power supplies.

3.6.6.5.1 The 0.25 to 50 volt programmable regulated voltage supply.- This supply shall provide thirty-two (32) different values of regulated voltage at variable spaced increments from 0.25 volts to 50 volts at I ampere maximum. Accuracy of any programmed value of voltage at one quarter maximum load current (250 milliampere) shall be ± 0.2 volt or ± 3 percent. The voltage shall remain within ± 0.1 volt or ± 2 percent of the value obtained by programming under the following conditions:

(a) plus or minus 10 percent change in line voltage

(b) 250 milliampere load current to no load current

The output voltage shall not exceed 0.1 volt peak-to-peak ripple under any output voltage programmed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.6.6.5.2 The 50 to 505 volt programmable regulated voltage supply.- This supply shall provide sixty-four (64) different values of regulated voltage at variable spaced increments from 50 volts to 505 volts at 50 milliamperes maximum between the voltages of 50 to 280 volts and 20 milliamperes maximum above 280 volts. Accuracy of programmed values of voltage between 50 and 280 volts at a test load current of 5 milliamperes shall be plus or minus 2 percent. Accuracy of programmed values of voltage above 280 volts at a test load current of 5 milliamperes shall be plus or minus 3 percent. The voltage shall remain within plus or minus 2 percent of the value obtained by programming under the following conditions:

(a) plus or minus 10 percent change in line voltage

(b) test load current to no load current

The output voltage shall have less than 1 percent peak-to-peak ripple under any output voltage programmed.

3.6.6.5.3 The programmable regulated current supply. - This supply shall provide twenty-four (24) different values of regulated current at variable spaced increments from 0.25 to 50 milliamperes at voltages up to 50 volts maximum. Accuracy of any programmed value of current at the half voltage point (25 volts) shall be plus or minus 2 percent. The current shall remain within plus or minus .05 milliamperes or plus or minus 2 percent of the value obtained by programming under the following conditions:

(a) plus or minus 10 percent change in line voltage

(b) change of load voltage from 25 volts to 0 volts.

The output current shall contain less than 0.4 percent peak-to-peak of ripple with the output load current programmed to 50 milliamperes.

3.6.7 Test circuit performance requirements.

3.6.7.1 h_{fe} (AC Beta) Test Circuit. - This circuit will provide the transistor under test with the proper connections to apply a programmed value of collector-to-base regulated voltage from the 0.25 to 50 volt supply, a programmed value of constant current between the base and emitter, a programmed value of 1 Kc base current modulation, and a method of extracting the AC component from the collector for use in measuring gain. When correctly programmed and used within the operating limits of the test set, a gain of from 5 to 990 should be measurable and may be compared to a programmed acceptance value of high limit and one of low limit for Go, No-Go indication.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.6.7.1.1 h_{fe} base drive. - This circuit shall generate a 1,000 cycle per second, plus or minus 2 percent, sine wave current signal. The output magnitude of this circuit shall be programmable in three current ranges: 0.2 microamperes peak-to-peak, 2.0 microamperes peak-topeak, or 20 microamperes peak-to-peak. Each range has an accuracy of plus or minus 2 percent.

3.6.7.2 h_{FE} (pulse) (DC Beta) Test Circuit. - This test circuit shall provide the transistor under test with the proper connections to apply a programmed value of collector-to-emitter voltage from the 0.25 to 50 volt supply through a programmed value of collector current measuring resistor, a programmed value of pulse current to the base, and a method of extracting the pulse component from the collector current measuring resistor for use in measuring gain. When correctly programmed and used within the operating limits of the test set, a gain of from 5 to 990 should be measurable and may be compared to a programmed acceptance value of high limit and one of low limit for Go, No-Go indication. The programmed values of collector current measuring resistor shall be 0.1 plus or minus 1 percent ohms, 0.2 plus or minus 1 percent ohms, 0.4 plus or minus 1 percent ohms, and 0.8 plus or minus 1 percent ohms.

3.6.7.2.1 h_{FE} (pulse) base drive. - The test circuit shall generate a pulse train of constant amplitude pulses 350 plus or minus 75 microseconds wide at a frequency of 35 plus or minus 10 cycles per second. Twelve (12) different programmed values of pulse current output shall be available for base drive:

1.0 amperes	±2%	100 milliamperes	±2%	10 milliamperes	±2%
500 milliamperes	±3%	50 milliamperes	±2%	5 milliamperes	±3%
250 milliamperes	±4%	25 milliamperes	±2%	2.5 milliamperes	±4%
125 milliamperes	±5%	12.5 milliamperes	±2%	1.25 milliamperes	±5%

3.6.7.3 h_{FE} (DC Beta) test circuit. - This test circuit shall provide the transistor under test with the proper connections to apply a programmed value of collector-to-emitter voltage from the 0.25 to 50 volt supply through a programmed value of collector current measuring resistor, a programmed value of direct current to the base, and a method of extracting the value of collector current from the collector current measuring resistor for use in measuring gain. When correctly programmed and used within the operating limits of the test set, a gain of from 5 to 990 should be measured and be compared to a programmed acceptance value of high limit and one of low limit for Go, No-Go indication.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.6.7.3.1 h_{FE} (DC) base drive.- This circuit shall develop a constant current for base drive which is programmable over twelve (12) different values:

1 milliampere	±3%	100 microamperes	±3%	10 microamperes	±3%
500 microamperes	±3%	50 microamperes	±3%	5 microamperes	±3%
250 microamperes	±3%	25 microamperes	±3%	2.5 microamperes	±3%
125 Microamperes	±3%	12.5 microamperes	±3%	1.25 microamperes	±3%

3.6.7.4 Current for voltage test circuit. - This test circuit shall provide the semiconductor junction under test with the proper connections to apply a programmed value of junction voltage from either the 0.25 to 50 volt supply of the 50 to 505 volt supply through a programmed value of series current measuring resistor, and method of extracting the value of current from the series resistor for use in measuring the current. When correctly programmed and used within the operating limits of the test set and in conjunction with the scaling amplifiers, a current of from 0.1 microamperes to 1 ampere should be measurable and may be compared to two programmed acceptance value limits for Go, No-Go indication. Four (4) programmed values of measuring resistor are available, as follows:

1 ohm ±1% - 100 ohms ±1% - 10K ohms ±1% and 100K ohms ±1%

3.6.7.5 Voltage for current test circuit. - This test circuit shall provide the semiconductor junction under test with the proper connections to apply a programmed value of series current from the programmable regulated current supply, a method of programming the 0.25 to 50 volt supply or the 50 to 505 volt supply in series with the current supply to extend its 50 volt capability for high voltage tests, and a method of extracting the value of junction voltage for use in measuring the voltage. Since the junction voltage can go to 505 volt, it would exceed the 9.9 volt maximum input to the projection digital voltmeter so a programmable resistive voltage divider of 1/10 plus or minus 1 percent or 1/100 plus or minus 1 percent shall be provided.

3.6.7.6 Zener impedance test circuit. - This test circuit makes use of of the voltage for a current test circuit of 3.6.7.5 and provides a means for injecting an AC signal into the programmable regulated current supply to make the current vary by the amount of 1.0 milliamperes plus or minus 0.1 milliamperes peak-to-peak at the line frequency, and a method of extracting the value of AC voltage resulting from the AC current from across the junction being tested for use in measuring the impedance. When correctly programmed and used within the operating limits of the test set and in conjunction with the scaling amplifiers, an impedance of from 3 ohms to 990 ohms should be measurable and may be compared to a programmed acceptance value high limit for Go, No-Go indication.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.6.8 Input power. - The primary circuits of the power supplies shall be capable of operation from 115 volts plus or minus 10 percent, or 230 volts plus or minus 10 percent, single phase at 60 cycles per second plus or minus 10 percent or 400 cycles per second plus or minus 10 percent. The transistor test set shall be capable of meeting the performance requirements of this specification with the above mentioned input voltages.

3.6.8.1 Power application. - Operation of the circuit breaker and power switch shall apply power to the transistor test set. If input voltage is 230 volts AC, no further operations shall be necessary. If the input voltage is 115 volts AC, the 115 SELECT button shall be activated to operate the voltage select circuit in the transistor test set.

3.6.8.2 Circuit breaker. - The primary circuits of the test set shall be protected against overloads in either the 115 volt or 230 volt mode of operation by a dual trip level circuit breaker.

3.6.9 Ground potential. - The transistor test set shall insure that all external parts of the test set are at ground potential at all times except for the diode test clips which receive momentary potentials of up to 505 volts during some test conditions.

3.7 Accessory kit. - The accessory case shall contain, as a kit, the following items:

		Part No.
(a)	Power input cable	8588147
(b)	In-circuit probe	8625490
(c)	Clip-lead adapter	8625521
(d)	Instruction manual	M8625349
(e)	Punched programmed cards	8625444
(f)	Card punch	8625592
(g)	Spare cards	8625444
(h)	Spare parts as listed on	8625348
(i)	Thermoelectric module	8625556

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities. - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the Contractor order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements. - Inspections shall be in accordance with Specification MIL-I-45208.

4.3 Initial production sample. - Prior to production, the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the Contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 Initial production sample failure. - Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Lot formation. - Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.1 Defective units and lots. - Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.5 Material. - Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.6 Inspection of components. - All positional and functional characteristics shown on drawings, or contained in notes on the drawings or references specifications, which indicate precision requirements and have a bearing on the functional accuracy and performance of transistor test set shall be inspected in accordance with the written inspection system required by Specification MIL-I-45208. These characteristics are defined as requirements effecting concentricity, squareness, parallelism, microfinishes 64 RUS or better, threads, identification marking and performance.

4.6.1 Sampling inspection. - Where quality control has been established on in-process manufacture, sampling inspection of components may be used in lieu of 100 percent inspection and shall be in accordance with Table I.

TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.6.2 Disposition of non-conforming product. - Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.7.1 Accuracy requirements. - Inspection equipment which incorporates features for measuring values or sizes established by product requirement shall be capable of measuring those values within an accuracy of ten percent of specified tolerance, except as specified in 4.7.2. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.7.2 Test equipment accuracy. - Test equipment utilized in this specification shall contain the following accuracies:

Item Description	Accuracy Requirements	
D-C Millimmeter	1.25 microampere range	±0.5%
	0.25 milliampere range	±0.5%
	1.0 milliampere range	±0.5%
	50 milliampere range	±0.5%
D-C Voltmeter	0.25 volt range	±0.5%
	2.5 volt range	±0.5%
	10 volt range	±0.5%
	50 volt range	±0.5%
	65 volt range	±0.5%
	500 volt range	±0.5%
RMS Voltmeter	Voltage range: 10 millivolt to 300 volts full scale in twelve ranges	
	Decibel range: -72 to +52 dbm in twelve ranges	
	Frequency response: 10 cycles to 7 megacycles	
	Input impedance: 10 megohms shunted by 30uuf on ranges from 0.001 volt to 0.300 volt, 10 megohms shunted by 15uuf on ranges from 1.0 volt to 300 volts	
	Accuracy: ±1% of full scale 50 cps to 800 kilocycles	
Oscilloscope	Passband: dc to 30 megacycles	
	Calibrated deflection factor: .05 to 20 volts per centimeter	
	Fastest sweep (magnified): 20 nanosecond per centimeter	
	Slowest sweep (calibrated): 5 second per centimeter, ±3% accuracy	
	Calibrated sweep magnifier: 5X	
	Cathode ray tube: accelerating voltage 10 kilovolts	
	Vertical scale: 4 centimeter	

Test Resistors			
	100 K ohm	±1%	2 watt
	10 K ohm	±1%	1/2 watt
	1 K ohm	±1%	2 watt
	200 ohm	±1%	10 watt
	1 ohm	±1%	1/2 watt

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.7.3 Inspection equipment design requirements. - The accuracy requirements for test equipment to perform environmental tests are specified in MIL-E-5272.

4.8 Acceptance inspection.

4.8.1 Classification of defects. - The classification of defects in Table II shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect any applicable requirement, and to reject individual nonconforming items. The AQL values specified pertain to each individual characteristic.

4.8.2 Disposition of rejected lots. - Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government informed of the corrective action taken, the lot shall be resubmitted for acceptance, as outlined in Standard MIL-STD-105.

Table II

Classification of Defects

Use Inspection Level II in Table II-A with Sampling Plan Table IV-A of Standard MIL-STD-105.

Critical: None defined.

Major: AQL 1.0%	Req't Par.	Q.A. Par.	Test Methods
101. Completeness of assembly	3.5	--	Visual
102. Projection digital voltmeter	3.6.4.1	4.9.2.3	Standard Measuring Equipment (SME)
103. Programmable decimal	3.6.4.1	4.9.2.3	SME
104. Off-Scale indicator	3.6.4.1	4.9.2.3	SME
105. Scaling amplifier	3.6.4.2	4.9.2.4	SME
106. Indicators	3.6.4.3	4.9.2.5	SME
107. Test routine	3.6.5	4.9.2.7	SME
108. Test set-up	3.6.6.1	4.9.2.7	SME
109. Test operation	3.6.6.2	4.9.2.5	SME
110. Programming	3.6.6.3	4.9.2.7	SME
111. Acceptance limits	3.6.6.4	4.9.2.3	SME
112. Programmable power supplies	3.6.6.5	4.9.2.6	SME

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

Table II

Classification of Defects (Cont'd)

Major: AQL 1.0%		Req't. Par.	Q.A. Par.	Test Methods
113.	0.25 to 50 volt regulated	3.6.6.5.1	4.9.2.6.1	SME
114.	50 to 505 volt regulated	3.6.6.5.2	4.9.2.6.2	SME
115.	Programmable current	3.6.6.5.3	4.9.2.6.3	SME
116.	Test circuits	3.6.7	4.9.2.7	SME
117.	h_{fe} (AC Beta)	3.6.7.1	4.9.2.7.1	SME
118.	h_{FE} (DC Beta pulsed)	3.6.7.2	4.9.2.7.2	SME
119.	h_{FE} (DC Beta)	3.6.7.3	4.9.2.7.3	SME
120.	Current for a voltage	3.6.7.4	4.9.2.7.4	SME
121.	Voltage for a current	3.6.7.5	4.9.2.7.5	SME
122.	Zener impedance	3.6.7.6	4.9.2.7.6	SME
123.	Nominal line voltage	3.6.8	4.9.2.8	SME
124.	Low line voltage	3.6.8	4.9.2.8	SME
125.	High line voltage	3.6.8	4.9.2.8	SME
126.	Power application	3.6.8.1	4.9.2.8	SME
127.	Ground potential	3.6.9	4.9.2.9	SME

Minor: None defined.

4.9 Acceptance inspection testing.

4.9.1 Environmental testing. - Environmental tests outlined below shall be performed on the initial production sample and one in every fifty; thereafter, the quantities specified above shall prevail provided the contractor continues manufacturing the item using the same processes and techniques used to produce the initial production sample. Any alteration in the manufacturing techniques or processes will require a new production sample and the sampling procedure shall begin again. The contractor shall be required to present a new production sample when any contract revision, determined by the contracting officer, indicates a change in performance requirements.

4.9.1.1 Low temperature.

4.9.1.1.1 Low temperature procedure I. - The transistor test set shall be placed within the test chamber and the chamber cooled to and maintained at a temperature of -20°C (-4°F) until stabilization of the equipment temperature is reached. While at this temperature, the equipment shall be tested to determine conformance to the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.9.1.1.2 Low temperature procedure II. - Following the test in 4.9.1.1.1, the temperature within the chamber shall be reduced to and maintained at a temperature of -62°C (-80°F) for 72 hours with the equipment de-energized. The temperature of the chamber shall then be raised slowly to the laboratory ambient temperature and the equipment shall meet the performance requirements of this specification.

4.9.1.1.3 High temperature. - The transistor test set shall be placed within the test chamber and the chamber temperature raised to +71°C (+160°F). The relative humidity within the chamber shall not be more than 15%. The equipment shall be maintained at this temperature for a period of 48 hours. At the conclusion of the exposure period, the temperature within the chamber shall be lowered to +50°C (+122°F). After equipment stabilization at +50°C (+122°F), the transistor test set shall be energized and operated at this temperature and shall meet the performance requirements of this specification. The chamber temperature shall then be lowered to the laboratory ambient temperature and a visual inspection performed to determine if any deterioration has occurred.

4.9.1.1.4 Altitude. - The transistor test set shall be placed in the test chamber and the internal chamber temperature reduced to -54°C (-65°F). The internal pressure of the chamber shall be reduced to 3.44 In. Hg. (50,000 ft. altitude). The equipment shall be maintained under these conditions for one hour. At the conclusion of this period, the internal chamber temperature shall be raised to -20°C (-4°F) and the internal pressure of the chamber shall be raised to 20.58 In. Hg. (10,000 ft. altitude). After the equipment has stabilized at the new temperature and pressure, the equipment shall be energized and shall meet the performance requirements of this specification. The chamber shall then be returned to laboratory ambient and pressure and inspected for visual damage.

4.9.1.1.5 Temperature shock. - The transistor test set shall first be placed within a test chamber wherein a temperature of 85 C (185 F) is maintained. The equipment shall be exposed to this temperature for a period of 4 hours, at the conclusion of which and within 5 minutes, the equipment shall be transferred to a chamber having an internal temperature of -40°C (-40°F). The equipment shall be subjected to this temperature for a period of 4 hours. This constitutes one cycle. The number of complete cycles shall be three. The duration of exposure at each extreme temperature shall not be less than that specified and may be extended to overnight exposure to prevent interruption of the transfer sequence. At the conclusion of the third cycle, the equipment shall be removed from the test chamber, returned to standard temperature, and within a period of 1 hour operated and inspected as directed in the test plan of Figure 1 and shall meet the performance requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.9.1.1.6 Shock. - The transistor test set, non-operating, with the covers on, shall be subjected to the shock conditions specified herein. The transistor test set shall be subjected to 18 impact shocks of 15g acceleration, each shock impulse having a time duration of 11 plus or minus 1 milliseconds. The maximum acceleration shall be reached in approximately 5-1/2 milliseconds. The shock shall be applied in the following directions:

(a) Vertically, 3 shocks in each direction

(b) Parallel to the major horizontal axis, 3 shocks in each direction

(c) Parallel to the minor horizontal axis, 3 shocks in each direction

The transistor test set shall be inspected to assure that it did not suffer damage and that it will provide the performance specified.

4.9.1.1.7 Vibration. - The transistor test set shall be non-operating with cover on and shall be vibrated to the following tests:

4.9.1.1.7.1 Vibration test 1. - The transistor test set shall be vibrated with the frequency varying between 10 and 33 cycles per second at a total excursion of 0.06 inch, the frequency varying uniformly from 10 to 33 cycles per second and returning to 10 cycles per second in approximately 1/2 minute.

4.9.1.1.7.2 Vibration test 2. - The transistor test set shall be vibrated with the frequency varying between 10 and 55 cycles per second at a total excursion of 0.03 inch, the frequency varying uniformly from 10 to 55 cycles per second and returning to 10 cycles per second in approximately 1 minute. The transistor test set shall be vibrated for 15 minutes in each of three mutually perpendicular planes for both Vibration Test 1 and Vibration Test 2.

4.9.1.1.7.3 Vibration test 3. - The transistor test set shall be vibrated for a period of 3 minutes at each of the four most severe resonant frequency points at an amplitude of 0.03 inch. The transistor test set shall then be inspected and shall meet the performance requirements of this specification to determine mechanical or electrical failures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.9.1.1.8 Humidity. - The transistor test set shall be placed in the test chamber and set up to simulate installed conditions. During the first 2 hour period, the temperature in the test chamber shall be raised from 28°C (82°F) to 71°C (160°F). The temperature of 71°C (160°F) and a relative humidity of 95 percent shall be maintained in the test chamber during the next 6 hour period. During the following 16 hour period, the temperature in the chamber shall be reduced at a uniform rate to 28°C (82°F) which constitutes one cycle. The cycle shall be repeated 10 times for a total test time of 240 hours. At the conclusion of the 240 hour period, the equipment shall be returned to laboratory environment and shall meet the requirements of this specification. Distilled or demineralized water having a pH value between 6.5 and 7.5 at 25°C (77°F) shall be used to obtain the desired humidity.

4.9.1.1.9 Sand and dust. - The front and rear fiberglas covers attached to empty test set center section (to simulate a transistor test set) shall be installed in the transistor test set transit case and placed in a test chamber equal to Specification MIL-C-9436. The internal temperature of the test chamber shall be maintained at 25°C (77°F) for a period of 6 hours with sand and dust velocity through the test chamber between 100 and 500 feet per minute. After 6 hours at the above conditions, the temperature shall be raised to and maintained at 71°C (160°F). These conditions shall be maintained for 6 hours. At the end of this test period, the equipment shall be removed and allowed to cool to room temperature. Accumulated dust shall be removed by brushing or wiping, care being taken to avoid introduction of additional dust into the equipment case. Under no circumstances shall dust be removed by either blast or vacuum cleaning. After cleaning, the inside of the equipment case shall be inspected for entry of sand and dust.

4.9.1.1.10 Splashproof. - The splashproof tests specified in Standard MIL-STD-108 will be performed on the front and rear fiberglas covers attached to an empty test set center section (to simulate a transistor test set). At the conclusion of the test after drying the outside of the equipment case, the inside shall be inspected for entry of water.

4.9.1.1.11 Salt spray. - Samples of the equipment structures, such as chassis, panels, or other mechanical assemblies shall be subjected to a salt spray (fog) for a period of 50 hours under controlled temperature and pH conditions as detailed in Standard FED-STD-151, except as otherwise specified herein. Following the 50 hour test period, the items tested shall be inspected for corrosion and pitting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.9.2 Examination.

4.9.2.1 Test conditions. - Unless otherwise specified, all tests shall be performed under the following conditions:

(a) Ambient Conditions; the tests shall be performed under factory or laboratory ambient conditions.

(b) Input Power; the input power shall be 115 volts plus or minus 2 percent 60 cycles per second plus or minus 2 percent.

(c) Warm-up; the equipment shall be turned ON and permitted to warm up for 5 minutes before any test is conducted.

4.9.2.2 Subassemblies. - The subassemblies listed in 3.6.1 shall meet the requirements of their respective specifications, when checked in the test set.

4.9.2.3 Projection digital voltmeter. - Testing the projection digital voltmeter to meet the requirements as specified in paragraph 3.6.4.1 shall be done in the following manner:

(a) The strap wire between TB2101-1 (NEG) and TB2102-1 (POS) shall be removed and a milliammeter to measure 1.0 milliampere with a reading accuracy of at least plus or minus 0.5 percent inserted in its place. No inspection test card is required. With the test set turned ON, the meter should read 1.0 milliampere plus or minus 1 percent. Turn OFF the test set, remove the meter and replace the strap wire.

(b) With test set turned ON, RESET to READY, switch to LEADS KNOWN, and insert inspection test card #1, Dwg. #8625444-1. With the AUTO-MANUAL-ADVANCE switch in MANUAL position, press the TEST button and then MANUAL ADVANCE into test position 5. While in test position 5, press the ACCEPTANCE LIMIT switch. The programmed acceptance value of 50 shall produce a projection readout (including the right hand zero) of from 470 to 530.

(c) MANUAL ADVANCE into test position 6 and press the ACCEPTANCE LIMIT switch. The programmed acceptance value of 96 shall produce a projection readout (including the right hand zero) of from 930 to 990.

(d) MANUAL ADVANCE into test position 8 and press the ACCEPTANCE LIMIT switch. The programmed acceptance value of 10 shall produce a projection readout (including the left hand zero and decimal point) of from 0.07 to 0.13.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(e) MANUAL ADVANCE into test position 9 and press the ACCEPTANCE LIMIT switch. The programmed acceptance value of 03 shall produce a projection readout (including the center decimal point) of from 0.0 to 0.6.

(f) RESET the test set to READY, remove inspection test card, and replace with inspection test card #2, Dwg. #8625444-2.

(g) MANUAL ADVANCE into test position 1 and press the ACCEPTANCE LIMIT switch. The OFF SCALE indicator shall light in addition to whatever two digit numerical readout appears. Neither decimal or left or right zero shall light.

(h) Repeat the above test after MANUAL ADVANCE into test position 2, then MANUAL ADVANCE through to test position 8, conduct the above test, then MANUAL ADVANCE to test position 9 and repeat the above test. In positions 1, 2, 8, and 9, the OFF SCALE indicator shall light. In position 8 and 9 the right hand zero shall light.

(i) RESET the test set to READY, and remove the inspection test card.

4.9.2.4 Scaling amplifier. - Testing the scaling amplifier to meet the requirements of paragraph 3.6.4.2 shall be done in the following manner:

(a) With the test set turned ON, using LEADS KNOWN, insert the inspection test card #1, Dwg. #8625444-1. Insert a 100K ohm plus or minus 1 percent 2 watt test resistor between pins 5 and 1 of J104 (between collector and emitter using clip-lead adapter).

(b) MANUAL ADVANCE into test position 6. The use of a scaling amplifier gain programmed to a value of 1,000 should produce a readout (including the right hand zero) from 460 to 540.

(c) Remove the one lead of the 100K ohm test resistor from pin 1 of J104 and connect to pin 3 of J104 (move from emitter clip to base clip using clip-lead adapter).

(d) MANUAL ADVANCE into test position 8. The use of a scaling amplifier gain programmed to a value of 100 should produce a readout (including the left hand zero and decimal point) of from 0.46 to 0.54.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(e) Remove the 100K ohm test resistor from between pins 5 and 3 of J104 (between collector and base using clip-lead adapter) and replace with a 10K ohm plus or minus 1 percent 1/2 watt in the same location.

(f) MANUAL ADVANCE into test position 9. The use of a scaling amplifier gain programmed to a value of 10 should produce a readout (including the center decimal point) of from 4.6 to 5.4.

(g) RESET the test set to READY and remove the inspection test card and test resistor.

4.9.2.5 Indicators. - Testing the indicators to meet the requirements of 3.6.4.3 shall be done in the following manner:

(a) With the test set turned ON, check the operation of the blue light. Using LEADS KNOWN, insert the inspection test card #2, Dwg. #8625444-2.

(b) Press RESET button and observe lighting of READY light.

(c) MANUAL ADVANCE through test positions 1 through 12 and observe the operation of the following in each test position:

(1) The twelve (12 digit projector shall indicate each TEST NUMBER in proper sequence.

(2) The NO-GO light shall light in test position 1.

(3) The GO light shall light in all test positions except 1.

(d) RESET the test set to READY. Place the input leads of an oscilloscope between pin 14 (common connection) of XDS1309 TEST NUMBER projector and ground to check for operating time. The oscilloscope controls should be set for 2 volts per centimeter vertical and 1 second per centimeter horizontal time base. Change AUTO-MANUAL-ADVANCE switch to the AUTO position and press the TEST button. As soon as the test sequence stops in position 1 and the NO-GO light lights, momentarily press the TEST DATA-ACCEPTANCE LIMITS switch to the right (which will release the "NO-GO" so the tests can continue). Within 6 seconds (plus any operator delay in operating the ACCEPTANCE LIMIT switch) the sequencing shall be complete and stop in position 12 with a GO indication.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(e) RESET the test set to READY, move switch to MANUAL position, remove the oscilloscope leads, remove the inspection test card and replace with the IN-CIRCUIT PNP test card, Dwg. #8625444-11 for In-circuit test of PNP transistors. Place the input leads of an oscilloscope across the contacts of capacitor C102 on the rear of the automatic lead selector - in-circuit checker chassis to check for operating time. The oscilloscope controls should be set for 10 volts per centimeter vertical and a 1/2 second per centimeter horizontal time base. Move switch to LEADS UNKNOWN and place a known good general purpose transistor of PNP polarity (such as a 2N404 or equivalent) into the TRANSISTOR test socket.

(f) Press the TEST button and within 3 seconds the leads shall have been found and a GO light shall have lit.

(g) RESET the test set to READY and place an external short between either the base and collector or the base and emitter of the transistor under test.

(h) Press the TEST button and within 3 seconds the SHORT light and the NO-GO light shall have lit.

(i) RESET the test set to READY and change the location of the external short to be between the collector and emitter of the transistor under test.

(j) Press the TEST button and within 3 seconds the NO GAIN light and the NO-GO light shall have lit.

(k) RESET the test set to READY, remove the external short from the transistor under test. Remove any one lead of the transistor under test from the test socket allowing the other two leads to remain in place.

(l) Press the TEST button and within 3 seconds the OPEN light and the NO-GO light shall have lit.

(m) RESET the test set to READY, remove the inspection test card, oscilloscope leads and test transistor.

4.9.2.6 Programmable power supplies. - Testing the programmable power supplies to meet the requirements of 3.6.6.5 shall be performed in the following manner:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.9.2.6.1 0.25 to 50 Volt Programmable Supply.

(a) With the test set turned ON, using LEADS KNOWN, insert the inspection test card #2, Dwg. #8625444-2. Place an external short between pins 5 and 3 of J104 (between collector and base using clip-lead adapter).

(b) MANUAL ADVANCE into test position 8. A high input impedance voltmeter to measure 0.25 volts with a reading accuracy of at least plus or minus 0.5 percent shall be placed across the lugs of TB2101-5 (POS) and TB2102-5 (NEG). Observe the voltmeter reading. The programmed value of 0.25 volts should read within plus or minus 0.2 volt while the RECHECK button is being held in (to provide a 250 milliampere load). When the RECHECK button is released (no-load to the supply) the voltage read on the voltmeter should not increase by more than 0.1 volt over the previous reading at a 250 milliampere load. When the line voltage is varied by plus or minus 10 percent while the RECHECK button is held in, the voltage read on the voltmeter should not change by more than plus or minus 0.1 volt of the previous reading.

(c) Remove and replace the voltmeter mentioned in (a) above on the same lugs with a voltmeter to measure 50 volts with a reading accuracy of at least plus or minus 0.5 percent. Place the input probe of an oscilloscope across the voltmeter and set the oscilloscope controls to measure ripple. Remove the external short mentioned in (a) above and replace with a power resistor of 200 ohms plus or minus 1 percent, with at least a 10 watt rating.

(d) MANUAL ADVANCE into test position 9 and observe the new voltmeter mentioned in (c) above while the RECHECK button is being held in. The programmed value of 50 volts should read within plus or minus 4 percent. The external 200 ohm test resistor has provided a 250 milliampere load. When the line voltage is varied by plus or minus 10 percent, the voltage read on the voltmeter should not change by more than plus or minus 2 percent of the previous reading. Observe the amplitude of the power supply ripple, as seen by the oscilloscope across the voltmeter while the RECHECK button is being held in. It should not exceed 0.1 volt peak-to-peak ripple, as seen by the oscilloscope across the voltmeter while the RECHECK button is being held in. It should not exceed 0.1 volt peak-to-peak ripple.

(e) Remove the external 200 ohm test resistor and the oscilloscope probe.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(f) While still in position 9, observe the voltmeter. The value of voltage previously measured in (d) above should not have changed by more than plus or minus 1 percent by going to no-load.

(g) RESET the test set to READY and remove the inspection test card and external voltmeter.

4.9.2.6.2 50 to 505 volt programmable supply.

(a) A high input impedance voltmeter to measure 65 volts with a reading accuracy of at least plus or minus 0.5 percent shall be placed between TB2101-4 (POS) and TB2102-4 (NEG). Place a 10K ohm plus or minus 1 percent 1/2 watt test resistor between pins 5 and 3 of J104 (between collector and base using clip-lead adapter). With the test set turned ON, using LEADS KNOWN, insert the inspection test card #3, Dwg. #8625444-3.

(b) MANUAL ADVANCE into test position 8 and observe the voltmeter mentioned in (a) above. The first programmed value of 62.5 volts should read within plus or minus 2 percent. The second programmed value of 50 volts should read within plus or minus 2 percent while the RECHECK button is being held in (to provide a 5 milliampere load). Remove the 10K ohm external test resistor and when the RECHECK button is being held in (no-load to the supply), the voltage read on the voltmeter should not change by more than plus or minus 2 percent of the previous reading at a 5 milliampere load for the second programmed value. Likewise, when the line voltage is varied by plus or minus 10 percent while the RECHECK button is held in, the voltage read on the voltmeter should not change by more than plus or minus 2 percent of the previous reading.

(c) Remove and replace the voltmeter specified in (a) above. Place on TB2101-3 (POS) and TB2102-4 (NEG) with a voltmeter to measure 495 volts with a reading accuracy of at least plus or minus 0.5 percent. Place a 100K ohm plus or minus 1 percent 2 watt external test resistor between pins 5 and 3 of J104 (between collector and base using clip-lead adapter). Place the input probe of an oscilloscope across the 100K ohm test resistor and set the oscilloscope controls to measure ripple.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(d) MANUAL ADVANCE into test position 9 and observe the new voltmeter mentioned in (c)on previous page. The programmed value 495 volts should be read within plus or minus 3 percent while the RECHECK button is being held in (to provide a 5 milliampere load). Observe the amplitude of the power supply ripple, as seen by the oscilloscope across the 100K test resistor while the RECHECK button is held in. It should not exceed 5.0 volt peak-to-peak ripple. Remove the 100K ohm external test resistor and when the RECHECK button is being held in (no-load to the supply), the voltage read on the voltmeter should not change by more than plus or minus 2 percent of the previous reading at a 5 milliampere load. Likewise, when the line voltage is varied by plus or minus 10 percent while the RECHECK button is held in, the voltage read on the voltmeter should not change by more than plus or minus 2 percent of the previous reading.

(e) RESET the test set to READY, remove the inspection test card, test resistor, voltmeter, and oscilloscope probe.

4.9.2.6.3 Programmable regulated current supply.

(a) With the test set turned OFF, switch to LEADS KNOWN and insert the inspection test card #4, Dwg. #8625444-4. The strap wire between TB2102-2 (NEG) and TB2101-2 (POS) shall be removed and a milliammeter to measure 50 milliamperes with a reading accuracy of at least plus or minus 0.5 percent in series with a 1K ohm plus or minus 1 percent 2 watt test resistor inserted in its place. Across the 10K ohm test resistor, place the probe of an oscilloscope and set the controls to measure ripple (scope ground lead at negative end). Turn the test set ON and MANUAL ADVANCE into test position 8. Place an external short between pins 5 and 3 J104 (between collector and base using clip load adapter). MANUAL ADVANCE into position 10.

(b) Press RECHECK button while still in position 10 and observe the milliammeter mentioned in (a) above. The programmed value of 50 milliamperes should read within plus or minus 2 percent. While still holding the RECHECK button in, vary the line voltage plus or minus 10 percent and observe that the current as read by the milliammeter does not change by more than plus or minus 1 percent of the previous reading. Observe the amplitude of the current supply ripple, as seen by the oscilloscope across the 1K test resistor while the RECHECK button is held in. It should not exceed 0.25 volts peak-to-peak ripple.

(c) While still in position 10, remove the oscilloscope probe and short out the 1K ohm test resistor, that is in series with the milliammeter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(d) Press RECHECK button while still in position 10 and observe on the milliammeter that the reading has not changed by more than plus or minus 1 percent of the reading first obtained in (b) on previous page. Remove external collector to base short.

(e) RESET the test set to READY, turn OFF the test set, remove the inspection test card, 1K ohm test resistor, and the milliammeter. Insert inspection test card #5, Dwg. #8625444-5, and place a milliammeter to measure .25 milliamperes with a reading accuracy of at least plus or minus 0.5 percent between TB2102-2 (NEG) and TB2101-2 (POS). Turn the test set ON and MANUAL ADVANCE to test position 8. Replace the external collector to base short specified in (b) on previous page, MANUAL ADVANCE into test position 11.

(f) Press RECHECK button while still in position 11 and observe the milliammeter mentioned in (e) above. The programmed value of .25 milliamperes should read within 0.05 milliamperes.

(g) RESET the test set to READY, turn the test set OFF, remove the inspection test card, the external short and milliammeter, replace the strap wire.

4.9.2.7 Test circuits. - Testing the test circuits to meet the requirements as specified in paragraph 3.6.7 shall be done in the following manner:

4.9.2.7.1 h_{fe} (AC Beta) Test Circuit.

(a) With the test set turned ON, using LEADS KNOWN, insert the inspection test card #5, Dwg. #8625444-5. Place a milliammeter to measure 0.25 milliamperes with a reading accuracy of at least plus or minus 0.5 percent in series with a 100K ohm plus or minus 1 percent 2 watt test resistor between pins 3 and 1 of J104 (between base (NEG) and emitter (POS) using clip-lead adapter). Across the 100K ohm test resistor, place the input connections of a 1 percent accurate RMS voltmeter with the meter ground lead on the emitter lead.

(b) MANUAL ADVANCE into test position 1 and observe the RMS voltmeter mentioned in (a) above while pressing the RECHECK button. The programmed value of 0.2 microamperes peak-to-peak AC base drive should read an RMS voltage of 7.07 millivolts plus 2 percent or minus 4 percent. Observe the value of current on the milliammeter while pressing the RECHECK button. The value of emitter-to-base current should not exceed 0.25 milliamperes for this PNP polarity test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(c) Replace the AC voltmeter with an oscilloscope input probe.

(d) While still in position 1, press the RECHECK button, and observe that the AC signal, as seen on the oscilloscope, is a 1,000 cycles per second plus or minus 2 percent signal.

(e) Remove the 100K ohm test resistor, oscilloscope input probe and milliammeter. Place a 10K ohm plus or minus 1 percent 1/2 watt test resistor between pins 3 and 5 of J104 (between base (POS) and collector (NEG) using clip-lead adapter) and across the 10K ohm test resistor, place the leads of a voltmeter to measure 10 volts with a reading accuracy of at least plus or minus 0.5 percent.

(f) MANUAL ADVANCE into test position 2 and observe the voltmeter while pressing the RECHECK button. The programmed value of base-to-collector voltage should read 10 volts plus or minus 4 percent for this PNP polarity test.

(g) Remove the inspection test card and reverse the polarity of the voltmeter. Replace with inspection test card #3, Dwg. #8625444-3. Press the RETEST button and observe that the voltage as programmed on this test card for NPN polarity is now 2.5 volts plus or minus 0.2 volts.

(h) RESET the test set to READY, and remove the inspection test card, the 10K ohm test resistor and the voltmeter. Replace with inspection test card #4, Dwg. #8625444-4. Place a milliammeter to measure 50 milliamperes with a reading accuracy of at least plus or minus 0.5 percent in series with a 1,000 ohm plus or minus 1 percent 2 watt test resistor between pins 3 and 1 of J104 (between base (NEG) and emitter (POS) using clip-lead adapter). Across the 1,000 ohm test resistor, place the connections of a 1 percent accurate RMS voltmeter with the meter ground lead on the emitter side.

(i) MANUAL ADVANCE into test position 2 and observe the RMS voltmeter mentioned in (h) above while pressing the RECHECK button. The programmed value of 20 microampere peak-to-peak AC base drive should read an RMS voltage of 7.07 millivolts plus 2 percent or minus 4 percent. Observe the value of current on the milliammeter while pressing the RECHECK button. The value of emitter-to-base current should not exceed 50 milliamperes.

(j) Remove the inspection test card and reverse the polarity of the milliammeter. Replace with inspection test card #3, Dwg. #8625444-3. Press the RETEST button and observe that the current for NPN polarity does not exceed 50 milliamperes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(k) RESET the test set to READY, remove the inspection test card, test resistor, milliammeter, and RMS voltmeter.

4.9.2.7.2 h_{FE} (pulse) (DC Beta) test circuit.

(a) With the test set turned ON, using LEADS KNOWN, insert the inspection test card #4, Dwg. #8625444-4. Place a 1 ohm plus or minus 1 percent 1/2 watt test resistor between pins 3 and 1 of J104 (between base and emitter using clip-lead adapter) across the 1 ohm test resistor place the input probe of an oscilloscope (ground lead on EMITTER clip-lead) and set the oscilloscope controls to measure pulse base drive. Place a DC voltmeter to measure 2.5 volts with a reading accuracy of at least plus or minus 0.5 percent between pins 1 and 5 of J104 (between emitter (POS) and collector (NEG) using clip-lead adapter).

(b) MANUAL ADVANCE into test position 3 and observe the oscilloscope mentioned in (a) above while pressing the RECHECK button. The programmed value of 1 ampere pulse base drive should read on the oscilloscope as a negative deflection equivalent to 1 volt plus or minus 3 percent for this PNP polarity test. The pulse width shall be observed on the oscilloscope as being between 275 microseconds and 425 microseconds at the 50 percent voltage point. The pulse repetition rate shall be observed on the oscilloscope as being between 25 cycles per second and 45 cycles per second. While pressing the RECHECK button, the voltmeter mentioned in (a) above should be observed. The programmed value of 2.5 volts collector-to-emitter voltage should read within plus or minus 0.2 volts for this PNP polarity test.

(c) Remove the inspection test card and reverse the polarity of the voltmeter. Replace with inspection test card #3, Dwg. #8625444-3. Press the RETEST button and observe that the oscilloscope deflection is still the same amplitude but now a positive deflection for NPN polarity and that the voltmeter reading for NPN polarity is the same value as above.

(d) RESET the test set to READY, remove the inspection test card, the voltmeter and test resistor. Replace with inspection test card #5, Dwg. #8625444-5, and a 200 ohm plus or minus 1 percent 10 watt test resistor in the same location as was the 1 ohm test resistor. The oscilloscope input probe remains as before.

(e) MANUAL ADVANCE into test position 4 and observe the oscilloscope while pressing the RECHECK button. The programmed value of 1.25 milliampere pulse base drive should read on the oscilloscope as a negative pulse deflection equivalent to 0.25 volts plus or minus 5 percent for this PNP polarity test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(f) RESET the test set to READY, remove the inspection test card, test resistor and oscilloscope input probe.

4.9.2.7.3 h_{FE} (DC Beta) test circuit.

(a) With the test set turned ON, using LEADS KNOWN, insert the inspection test card #4, Dwg. #8625444-4. Place a DC voltmeter to measure 2.5 volts with a reading accuracy of at least plus or minus 0.5 percent between pins 1 and 5 of J104 (between emitter (POS) and collector (NEG) using clip-lead adapter).

(b) MANUAL ADVANCE into test position 5. Place a milliammeter to measure 1 milliampere with a reading accuracy of at least plus or minus 0.5 percent between pins 3 and 1 of J104 (between base (NEG) and emitter (POS) using clip-lead adapter). Observe the milliammeter and voltmeter while pressing the RECHECK button. The programmed value of 1 milliampere base drive should read between 0.97 milliamperes and 1.03 milliamperes for this PNP polarity test. The programmed value of 2.5 volts collector-to-emitter voltage should read within plus or minus 0.2 volts of 2.5 volts for this PNP polarity test.

(c) Remove the inspection test card and reverse the polarity of both the voltmeter and milliammeter. Replace with inspection test card #3, Dwg. #8625444-3. Press the RETEST button and observe that the voltmeter and milliammeter readings for NPN polarity are the same values.

(d) RESET the test set to READY, remove the inspection test card, the milliammeter and the voltmeter. Replace with inspection test card #5, Dwg. #8625444-5. Place a voltmeter to measure 12.5 millivolts with a reading accuracy of at least plus or minus 0.5 percent between pins 3 and 1 of J104 (between base (NEG) and emitter (POS) using clip-lead adapter). Place a 10K ohm plus or minus 1 percent 1/2 watt test resistor in parallel with the voltmeter between pins 3 and 1 of J104 (between base and emitter using clip lead adapter).

(e) MANUAL ADVANCE into test position 6 and observe the projection digital voltmeter readout. The programmed value of 10 volts emitter-to-collector voltage across the 10K ohm test resistor in series with the programmed internal measurement resistor of 80 ohms and the programmed scaling amplifier gain of X100 should produce a readout from 75 to 85. While pressing the RECHECK button, observe the microameter mentioned in (c) above. The programmed value of 1.25 microamperes base drive current should read within plus or minus 3 percent of 12.5 millivolts across the 10K test resistor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(f) RESET the test set to READY, remove the inspection test card, the test resistor and microameter.

4.9.2.7.4 Current for a voltage test circuit.

(a) With the test set turned ON, using LEADS KNOWN, insert the inspection test card #6, Dwg. #8625444-6. Place an external short between pins 5 and 3 of J104 (between collector and base using clip-lead adapter).

(b) MANUAL ADVANCE into test position 7 and observe the projection digital voltmeter presentation. The programmed value of 5 volts across the programmed internal measuring resistor of 100 ohms should produce a reading of from 47 to 53.

(c) RESET the test set to READY, remove the inspection test card and external short.

(d) The test routine of section 4.9.2.4 and 4.9.2.6 has exercised the other capabilities of this test circuit. Check results of the tests in 4.9.2.4 (d) and (f), 4.9.2.6.1 (b) and (d), and 4.9.2.6.2 (b) and (d), and if they are acceptable, then this test circuit may be considered acceptable.

4.9.2.7.5 Voltage for a current test circuit.

(a) With the test set turned ON, using LEADS KNOWN, insert the inspection test card #6, Dwg. #8625444-6.

(b) MANUAL ADVANCE into test position 11 and observe the projection digital voltmeter presentation. The programmed value of 0.75 milliamperes (using the current supply and voltage supply in series) through the internal test resistor produces a voltage which is divided 10:1 and is presented as a projection readout of from 71 to 79.

(c) RESET the test set to READY, remove the inspection test card.

(d) The test routine of section 4.9.2.6.3 has exercised the other capabilities of this test circuit. Check results of the tests in 4.9.2.6.3 (b) and (f) and if they are acceptable, then this test circuit may be considered acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.9.2.7.6 Zener impedance test circuit.

(a) With the test set turned ON, using LEADS KNOWN, insert the inspection test card #7, Dwg. #8625444-7. Place a 200 ohm plus or minus 1 percent 10 watt resistor between pins 3 and 5 of J104 (between base and collector using clip-lead adapter).

(b) MANUAL ADVANCE into test position 12 and observe the projection digital voltmeter presentation. The readout shall be from 160 to 240.

(c) RESET the test set to READY, remove the inspection test card, and test resistor.

4.9.2.8 Supply line variation test. - To test the input power requirement as specified in paragraph 3.6.8, the testing of the input power shall be performed in the following manner:

(a) Conduct the tests specified in 4.9.2.7.3 (e) to establish the readings at nominal input conditions. During power turn-on, observe use of 115 SELECT button.

(b) Change the line voltage to 103.5 volts. Observe the results of repeating the test 4.9.2.7.3 (e). Deviations shall not exceed previously specified accuracies for each parameter.

(c) Change the line voltage to 126.5 volts. Observe the results of repeating the test 4.9.2.7.3 (e). Deviations shall not exceed previously specified accuracies for each parameter.

(d) With the line voltage still at 126.5 volts, change the line frequency to 440 cycles per second. Observe the results of repeating the test in 4.9.2.7.3 (e). Deviations shall not exceed previously specified accuracies for each parameter.

(e) Change the line voltage to 103.5 volts and the frequency to 360 cycles per second. Observe the results of repeating the test in 4.9.2.7.3 (e). Deviations shall not exceed previously specified accuracies for each parameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

(f) RESET the test set to READY, turn power OFF by throwing the circuit breaker. Change line voltage to 230 volts, and frequency to 60 cycles per second. Turn power ON by throwing the circuit breaker and observe turn-on characteristics with and without use of 115 SELECT button. Conduct the test specified in 4.9.2.7.3 (e). Deviations shall not exceed previously specified accuracies for each parameter.

(g) Change the line voltage to 207 volts. Observe the results of repeating the test in 4.9.2.7.3 (e). Deviations shall not exceed previously specified accuracies for each parameter.

(h) Change the line voltage to 253 volts. Observe the results of repeating the test in 4.9.2.7.3 (e). Deviations shall not exceed previously specified accuracies for each parameter.

(i) With the line voltage still at 253 volts, change the line frequency to 440 cycles per second. Observe the results of repeating the test in 4.9.2.7.3 (e). Deviations shall not exceed previously specified accuracies for each parameter.

(j) Change the line voltage to 207 volts and the frequency to 360 cycles per second. Observe the results of repeating the test in 4.9.2.7.3 (e). Deviations shall not exceed previously specified accuracies for each parameter.

4.9.2.9 Ground test. - To meet the ground test requirements as specified in paragraph 3.6.9, perform the following test:

(a) With the test set turned ON, and the connector covers installed on the AUXILIARY POWER and REMOTE-ADAPTER plugs, insert inspection test card #3, Dwg. #8625444-3 to program the high voltage (495 volts). MANUAL ADVANCE to test position 9. Connect one lead of a voltmeter capable of measuring 10 volts to the ground shell of the input power cord. Using the other lead of the meter, probe or touch each exposed metal part of the test set not covered with a non-conductive coating such as paint, and check for voltmeter deflection. Do not connect the meter lead to the diode test clips. The test set is considered acceptable, if no meter deflection in excess of 1 volt is obtained.

(b) RESET the test set to READY, and remove the inspection test card and voltmeter.

4.9.2.10 Quality assurance provisions for packaging, packing and marking. - Quality assurance provisions for packaging, packing and marking shall be in accordance with Specification MIL-P-14232.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. - Preservation, packaging and packing shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8625349. The levels of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The transistor test set is an instrument used to test transistor and semiconductor devices which fall into the design limits of the instrument.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Selection of applicable level of preservation, packaging and packing

(c) Applicable stock number

(d) Provisions for initial production samples including data of delivery (See 3.1).

6.3 Definitions.

6.3.1 Standard measuring equipment (SME). - Standard measuring equipment (SME) is defined as the common type measuring devices which are usually stocked by commercial supply houses ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.3.2 Tactile. - Skillful manipulation or sensing.

6.3.3 Initial production sample. - A model in its final mechanical and electrical form of final production design made by production methods, tools, jigs and fixtures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any invention that may in any way be related thereto.

Custodian:

Army - MU

Preparing activity:

Army - MU

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.